

Stagnation in Physics, Inspiration from the Skies: Rethinking Science in the Age of UAP

Karl Svozil, ORCID ID: 0000-0001-6554-2802

Dedicated to the memory of Air Chief Marshal Hugh Caswall Tremenheere Dowding, 1st Baron Dowding, president of the Flying Saucer Club and allegedly member of the Fairy Investigation Society and the Ghost Club.

When confronted with the alleged impossibility of the proposed Chain Home radar system 1937, Dowding demanded:

“If the scientists say it's not possible, get me another set of scientists.”

My Own Experiences and Motivation

- ▶ My main driving force was I want to go to the stars fast—superluminal interstellar space travel.
- ▶ I am no experimenter. I had dreams of nuclear armageddon, somehow in a UFO context. I maybe saw a Tic Tac-shaped object in the cloudy skies in Upper Austria during swimming in a lake (unreliable, maybe mundane).
- ▶ I am primarily interested in the propulsion technique(s)—which I would like to call “**The Drive**”—that would fast-forward physics into a new era, thereby skipping maybe thousands of years of scientific (r)evolution(s).
- ▶ Secondly, I am interested in supplementary techniques such as transparent interfaces (“craft suite”) to the outside medium.
- ▶ My first encounter with the Condon Report was at the UCB library in 1984 during my stay at LBL; it was not available to me in Austria. I got the message—“no worries, almost nothing to see here; if true, relax”—by reading Condon’s executive summary.

Disclosure & Working Hypothesis

- ▶ **Defining Disclosure:** In the NHI context, “disclosure” signifies the official recognition of a non-human intelligence (NHI) presence on Earth by a globally significant authority (e.g., the President of the United States or the UN Secretary-General).
- ▶ **My Working Hypothesis:** I operate under the hypothesis that these intelligences are extraterrestrial, may operate from bases on or near Earth, and that human civilization could unknowingly be part of a vast galactic hypercivilization.
- ▶ **Important Caveat:** Many of the following considerations remain valid, even if this specific hypothesis is incorrect.

A Proposed Pragmatic Framework for UAP Inquiry

- ▶ **Stagnation and divergence in Physics:** Modern physics, despite its successes and erudition “piled higher & deeper,” lacks a unified theory, with General Relativity and Quantum Mechanics remaining conceptually distinct and lacks unity. Advanced science could help!
- ▶ **What About Extraordinary Claims?** Russell’s Teapot illustrates that the burden of proof lies on the claimant. Carl Sagan’s standard, “Extraordinary claims require extraordinary evidence,” is a core tenet of skepticism. In Bayesian terms, a claim with low prior probability requires strong evidence. However, the **immense potential utility of UAP findings for physics and other sciences could justify investigation even with low initial odds.**
- ▶ **Pragmatic Approach:** Adopting Hans Vaihinger’s philosophy of “As If” allows for a pragmatic “what if” approach. This is a common scientific practice to explore possibilities in the face of uncertainty.

The “Image” of a Physical Theory

- ▶ **Hertz’s “Image” of Thought:** The conceptual core of a physical theory.

“We form for ourselves images or symbols of external objects; and the form which we give them is such that the necessary consequents of the images in thought are always the images of the necessary consequents in nature of the things pictured. . . . we do not know, nor have we any means of knowing, whether our conceptions of things are in conformity with them in any other than this one fundamental respect.”

- ▶ **No Ontological Convergence:** The historical succession of images does not reveal a steady progression towards a single, ultimate reality.

- ▶ *Example (Gravity):* Epicycles (Ptolemaus) →
Force (Newton) →
Spacetime Curvature (Einstein) →
What comes next??

Evaluating and Understanding New Images

- ▶ **Lakatos' Research Programs:** It is impossible to know in advance whether a new image will be scientifically progressive or degenerative. Degenerative programs can seem more successful initially due to a protective belt of auxiliary theories, while a new image is still underdeveloped, with scarce empirical support.
 - ▶ *Economic Metaphor:* Like predicting markets without insider information, forecasting a theory's success is fraught with uncertainty.
- ▶ **Quantitative Clarke's Principle:** Without explanation (an "Alien or NHI Prometheus"), it is impossible to decipher a UAP technology that is more than one image ahead of our own.
 - ▶ *Example:* Post-WWII German technology was easily absorbed because it posed no great leap in the underlying physical image and Nazi experts, through *Paperclip*, were readily available to explain it.

Methodology for UAP Investigation

- ▶ **Caution Towards Current Explanations:** Do not uncritically accept pronouncements from current physicists on UAP.
 - ▶ They risk overconfidence (Dunning-Kruger effect) and may be ego-invested in their own pet theories.
 - ▶ *Example:* Seemingly impossible acceleration is no issue if the underlying image allows for inertia/mass to be reduced or nullified.
- ▶ **What Can Be Learned:**
 - ▶ **Recognize Possibility:** UAP sightings, regardless of origin, demonstrate physical capacities achievable in our universe.
 - ▶ **Embrace Open Inquiry:** We must pursue multiple research directions (Feyerabend) and expand knowledge through systematic trial and error (Popper).

What is Inertia?

- ▶ **The Core Tenet of General Relativity:** Gravity and inertial mass are fundamentally linked by the Equivalence Principle, the cornerstone of GR.
- ▶ **The Key Question:** What if we could manipulate this connection? If we could reduce or even invert inertial and gravitational mass, we could potentially mimic many of the observed flight characteristics of UAP.
- ▶ **An Exotic Example—The Inerter Analogy:** The “inerter,” a device from control theory used in Formula 1, acts as a mechanical motion attenuator. An inverse inerter could produce “negative impedance”, a sort of “negative mass”.

My Favorite Speculative Pet Theories: Manipulating Mass & Fractal Spacetime

- ▶ Currently inertial mass is explained twofold:
 - ▶ by the Higgs Mechanism, and
 - ▶ by the field (quanta such as gluons & photons) surrounding color & electromagnetic charges.
 - ▶ So, changing any of those affects inertia.
- ▶ **Casimir & Scharnhorst Effects:**
 - ▶ Mass reduction between parallel conducting plates via the Casimir effect (less radiative corrections).
 - ▶ Scharnhorst effect: apparent faster-than-vacuum light speed between the plates due to reduced vacuum polarization.
- ▶ **“Anti-gravity” in Fractal Spacetime:**
 - ▶ If spacetime has “holes” or a fractal structure, a craft could traverse “less” spacetime, “hopping” through gaps, during geodesic motion.
 - ▶ Could such a structure emerge from nonlinear physical processes?

A History of Anti-Gravity (AG) Efforts: Private Funding

- ▶ **Known Efforts (from late 1940s):**
 - ▶ Research funded by private donors Roger Babson and Agnew Bahnson.
 - ▶ Articles and book chapters by Kaiser & Rickles.
- ▶ **Previously Undisclosed Effort:**
 - ▶ A private initiative funded by the owner of Church's Texas Chicken.
 - ▶ **Personnel Involved:** Yakir Aharonov, with brief involvement from Danny Greenberger. Hal Puthoff's name has also been associated with this sponsor.

USAF Involvement and the “Going Dark”

- ▶ **USAF and Lab Research:** Numerous public and private labs, notably connected to the US Air Force, were involved in gravity research.
- ▶ **Witness Testimony (AIP Oral Histories):**
 - ▶ **Joshua Goldberg:** Recalls being hired at Wright-Patterson AFB because an officer believed an “anti-gravity device” was the next big thing. Confirms USAF was funding general relativity research in the late 50s/early 60s.
 - ▶ **Louis Witten:** Mentions the Air Force’s interest in AG in the mid-1950s and confirms receiving a “cost-plus” research contract directly from the Air Force while at RIAS.
- ▶ Private experiences and speculations about USG efforts by Australian government physicist **Harry Oliver Turner**.
- ▶ **Did Antigravity Legacy Research Go Dark? The Mansfield Amendment of 1969** required military-funded research to have a direct or apparent relationship to a specific military function.

Caution: The “Quantum” Gray Box

- ▶ The term “quantum” is often used as a hocus pocus box into which all of one’s esoteric wishes are injected.
- ▶ This frequently amounts to wishful thinking, without a solid knowledge of quantum mechanics proper.
- ▶ **Example: Misuse of “Entanglement”**
 - ▶ Entanglement is often co-opted to explain psychic (psy) phenomena.
 - ▶ This popular usage ignores its most fundamental property: the loss of **individual** properties among the entangled constituents for the sake of encoding **relational** properties.

Legacy Programs: Definitions and Custodianship

▶ Defining the “Legacy Program”:

- ▶ A non-public, highly secret, and deeply compartmentalized operation.
- ▶ A “member” is defined as a person with direct, physical access to Non-Human Intelligence (NHI) created craft, biological remains, or materials.
- ▶ To date, James T. Lacatski is the only public figure who has admitted to being part of such a program.

▶ The “Saucer Portfolio” and its Gatekeepers:

- ▶ According to multiple accounts, this portfolio is managed by the CIA’s **Directorate of Science and Technology (DS&T)**.
- ▶ Physical retrieval operations are reportedly handled by the CIA’s **Office of Global Access (OGA)**, whose mission is to “deliver intelligence from the most challenging denied areas.”
- ▶ This operational structure immediately makes all previous and current Deputy Directors of these departments the primary gatekeepers of the legacy UFO programs.

Historical Precedent (Marshall Chadwell, OSI, 1952)

- ▶ In a report on “British Activity in the field of ‘Unidentified Flying Objects’,” Marshall Chadwell wrote (with my emphasis):
*“During the day, a ‘perfect flying saucer’ was seen by these officials as well as RAF pilots. This is disturbing to Jones (a WWII radar expert, the “British Robertson”) because he realizes **that the creation of the correction of public opinion is part of his responsibilities.**”*
- ▶ Chadwell is seen here to align British and US efforts.
- ▶ Any questions who is the master and who the aligned?
- ▶ If already Britain is treated that way, what about, say, the WWI & WWII losers Germany or Austria?

Hypothetical Scenario: Another UAP Crash in Magenta, Italy

- ▶ **The Response:** An extraction operation would almost certainly be initiated within a couple of hours and completed within a few days, thereby bypassing unreliable and unforeseeable civilian political channels.
- ▶ **The Method:** The process would be managed at the military and security enforcement levels, likely involving a CIA-led OGA extraction team with a component from the Joint Special Operations Command (JSOC), with pre-existing local liaisons, and standardized protocols for processing “foreign (mundane) craft.”
- ▶ **A Stark Warning to Civilian Witnesses:** I highly recommend all civilian witnesses, for the sake of survival, to leave ASAP, and get out of anybody’s way.
- ▶ **The Resulting Questions:** In a context where official processes are opaque and civilian access is actively discouraged, what is left for civil society and science proper to do? Can broader research basis proceed when the subject is controlled by secretive, unaccountable entities?

The Geopolitical Status

- ▶ **A Permanent State of Emergency:** Since WWII, the USA in particular has existed in a state of permanent UFO-related emergency (the “Ausnahmezustand” described by Carl Schmitt and Giorgio Agamben), where normal rules are suspended.
- ▶ **Thucydides’ Melian Dialogue as a Metaphor:** The empire takes what it needs, while the colonies (or allies) suffer what they must.
- ▶ Again we may ask: Where does this leave Europe and other international partners?
- ▶ I guess they are seen as potential adversaries. It can be assumed that the same can be said of Russian and Chinese efforts.

A Potential Core Dilemma of Disclosure: Nonexistence of Autonomy

- ▶ **Autonomy:** Do we, as a society or even as a species, possess the autonomy to make a disclosure decision? Or is that choice no longer ours? Are NHIs prohibiting disclosure?
- ▶ **The Containment Hypothesis:** We may present a potential threat to Aliens. If we could effortlessly travel to them, we might cause inconveniences. Therefore, they might share some goodies but never the secret to their propulsion and travel—**The Drive**.

A Proposed Path Forward (If We Can Choose)

- ▶ **Utopian Strategy:** Adopt an Open Access approach, fostering dialogue between all relevant parties, including the US, Russia, and China. This may also involve consultation with NHI presence, if feasible.
- ▶ **Guiding Principle wrt Fear of Disclosure:** The Enlightenment sought to eliminate fear with rationality. Duke Dracula was overcome by science.
- ▶ A quote by Immanuel Kant, 1784:
Enlightenment is man's emergence from his self-imposed immaturity (selbstverschuldeter Unmündigkeit). . . . Sapere Aude! [Latin translated: Dare to know, from Horace]. Have courage to use your own mind! Thus is the motto of Enlightenment.

Thank You for Your Attention